

PRAGA.SV

CT $\rightarrow$ (Read only) $\sim$ stores the encrypted input [0] $\sim$ length

PT $\rightarrow$ (WRITE ONLY) $\sim$ stores the decrypted output [1..n] $\sim$ enc. byts

pseudocode

i=0

j=0

message_length = ciphertext[0]

for k=1 to msg-length:

i = (i+1) mod 256

j = (j + s[i]) mod 256

swap(s[i], s[j])

pad[k] = s(s[i] + s[j]) mod 256

plaintext[0] = message_length

for k=1 to message_length:

plain-text = pad[k] XOR ciphertext[k]

section 1: Read the message length from ciphertext[0]

section 2: generate the keystream pad[k] for k=1 to msg-length

* pad[k] $\rightarrow$ immediate register datapath

pad[k] = s(s[i] + s[j]) mod 256

section 3: decrypt XORING ciphertext CT[k] with keystream pad[k]

PT[k] = pad[k] XOR CT[k]

KSA $\rightarrow$ sets up S using the secret key

PRGA $\rightarrow$ uses S to generate PAD[k] (keystream)

PT[k] $\rightarrow$ PAD[k] XOR CT[k]

SAMPLE TRACE

SMGM $\rightarrow$ 00 01 02 03 04 05 06 07 08 09 10 11 12 13 14 15

REF $\rightarrow$ B4 04 2B E5 49 0A 90 9A E4 17 F4 10 3A 35 13 77

(After KSA)*

key = 'h6d03/3c'
0 1 2

$\rightarrow$ $\begin{cases} \text{key [23:16]} \\ \text{key [15:8]} \\ \text{key [7:0]} \end{cases}$

CT = [03, CT1, CT2, CT3]

$(i=0, j=0)$

version 1

Read Message Length: $CT[0] = 03$

WRITE PLAINTEXT $PT[0] = 03$

version 2

Loop ($k=1$ to 3) $\rightarrow$

(k=1)

$i = (0+1) \bmod 256 \rightarrow (i=1)$ 'h04

$j = (j + \sqrt{CT[i]}) \bmod 255 \rightarrow j = (0 + \sqrt{CT[1]}) = (j=4)$

$\text{swap}(\sqrt{CT[i]}, \sqrt{CT[j]}) \rightarrow (\sqrt{CT[1]}, \sqrt{CT[4]})$

$\text{pat}[1] = \sqrt{(\sqrt{CT[1]} + \sqrt{CT[4]}) \bmod 255} \rightarrow$
'h49 'h04

$PT[1] =$ -

(k=2)

$i = (1+1) \bmod 256 \rightarrow (i=2)$ 'h2B ('d43)

$j = (4 + \sqrt{CT[2]}) \bmod 255 \rightarrow j = (4 + \sqrt{CT[4]}) = (47)$

$\text{swap}(\sqrt{CT[i]}, \sqrt{CT[j]}) \rightarrow (\sqrt{CT[2]}, \sqrt{CT[47]})$

$\text{pat}[2] = \sqrt{(\sqrt{CT[2]} + \sqrt{CT[47]}) \bmod 255} \rightarrow$
? ?

$PT[2] = \text{pat}[2] \text{ XOR } CT[2]$

(k=3)

$i = (2+1) \bmod 256 \rightarrow (i=3)$ 'h55 ('d229)

$j = (47 + \sqrt{CT[3]}) \bmod 255 \rightarrow j = (47 + \sqrt{CT[47]}) = (j=276)$ -(j=20)

$\text{swap}(\sqrt{CT[i]}, \sqrt{CT[j]}) \rightarrow (\sqrt{CT[3]}, \sqrt{CT[20]})$

$\text{pat}[3] = \sqrt{(\sqrt{CT[3]} + \sqrt{CT[20]}) \bmod 255} \rightarrow$
? ?

$PT[3] = \text{pat}[3] \text{ XOR } CT[3]$

v0: IOLf

v1: READ_CT_Length

v2: WRITE_PT_Length

v3: READ_CT

v4: READ_CT

v5: WRITE_CT

v6: WRITE_CT

v7: READ_PT

v8: WRITE_PT